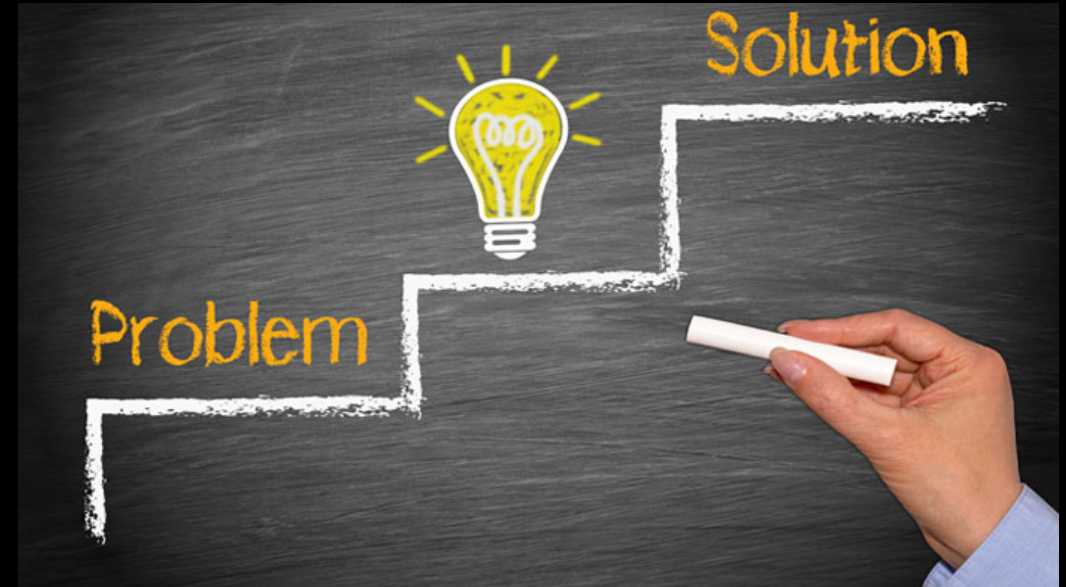


Automated Network Design and Deployment

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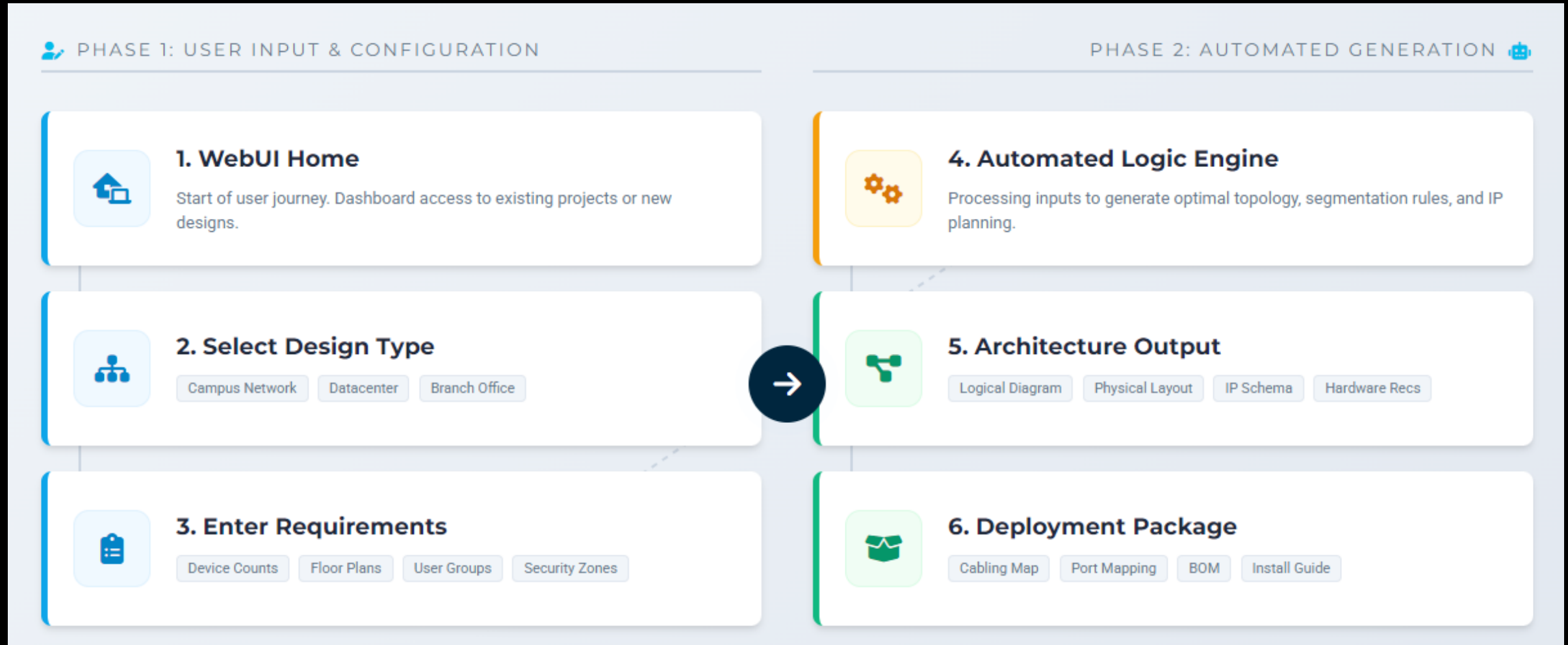
Project Details

Theme/Title: Automated Network Design and Deployment

- Objective/Abstract:

- This project aims to develop an intelligent, automated system capable of designing and guiding the deployment of a complete campus network without requiring expert networking skills. Students will build a software tool that collects essential inputs from non-technical users—such as building details, device counts, user categories, and security requirements—and automatically generates a fully validated HPE Aruba Networking-based campus network design.
- The system will recommend suitable HPE switches, gateways, routers, backbone connectivity, segmentation policies, IP addressing schemes, and PoE needs for devices such as IP cameras, VoIP phones, printers, desktops, and other wired or wireless client endpoints. Wi-Fi is treated simply as one of the many end-client access types, without requiring any advanced RF design or specialized wireless planning.
- The tool will also generate easy-to-follow deployment materials including network topology diagrams, cabling maps, switch-port mappings, configuration workflows, and bill-of-materials. To ensure technical accuracy, the solution incorporates an AI-powered “Grounded Design Copilot” that retrieves guidance from Aruba Validated Solution Guides, AOS-CX documentation, AOS-10 gateway references, and Aruba Central workflows, ensuring all recommendations align with industry best practices.
- The end goal is to enable non-technical staff to successfully deploy an enterprise-grade campus network, while giving students hands-on experience in automation, network architecture, design logic, and real-world implementation practices.

Architecture



Input Requirement & Configuration

DATA CAPTURE SCOPE

Site Structure

Physical hierarchy definition including Buildings, Floors, Rooms, and MDF/IDF locations with cable run distance constraints.

Device Inventory

Hardware counts and types: Switches, Access Points, IP Phones, Cameras, IoT Sensors, and physical server connections.

User Groups & Traffic

Classification of network consumers: Staff, Students, Guests, IoT devices, and OT/Industrial systems.

Sensitive Zones

Security segmentation requirements for compliance: Research Labs, Finance, Healthcare (HIPAA), and PCI zones.

Redundancy & HA

Availability requirements: High Availability pairs, dual uplink paths, power redundancy, and failover scenarios.

INPUT METHODS & VALIDATION

Data Ingestion Channels

Guided Web Wizard

CSV Bulk Import

REST API

Flexible input options allow for manual entry via wizard or bulk ingestion from existing CMDBs.

Intelligent Validation

Cabling Reach Check

PoE Budget Calc

Port Capacity

Real-time logic engine checks for physical constraints (e.g., >100m copper runs) and power oversubscription.

Configuration Presets

Campus Standard

DC Spine-Leaf

Branch Office

Start quickly with industry-standard templates that pre-populate best-practice configurations.

Confidential | Authorized 4

Automated Logic & Architecture Generation



Topology Synthesis

- Selects optimal model: Core/Dist/Access vs. Spine-Leaf based on scale.
- Determines physical stacking strategy and uplink consolidation.
- Evaluates ring vs. mesh topology for distribution layers.



Segmentation Logic

- Auto-generates VLAN ID mapping and VRF assignments.
- Defines TrustSec tags or ACL templates for microsegmentation.
- Configures guest isolation policies and IoT separation rules.



Intelligent Addressing

- Calculates efficient IPv4/IPv6 subnets based on user density.
- Generates route summarization plans for Core layers.
- Allocates DHCP scopes, gateway IPs, and management loops.



Capacity Planning

- Validates PoE budgets against device inventory (AP/Camera load).
- Calculates uplink bandwidth and oversubscription ratios (e.g., 20:1).
- Recommends link aggregation groups (LAG/LACP) where needed.



Security & Isolation

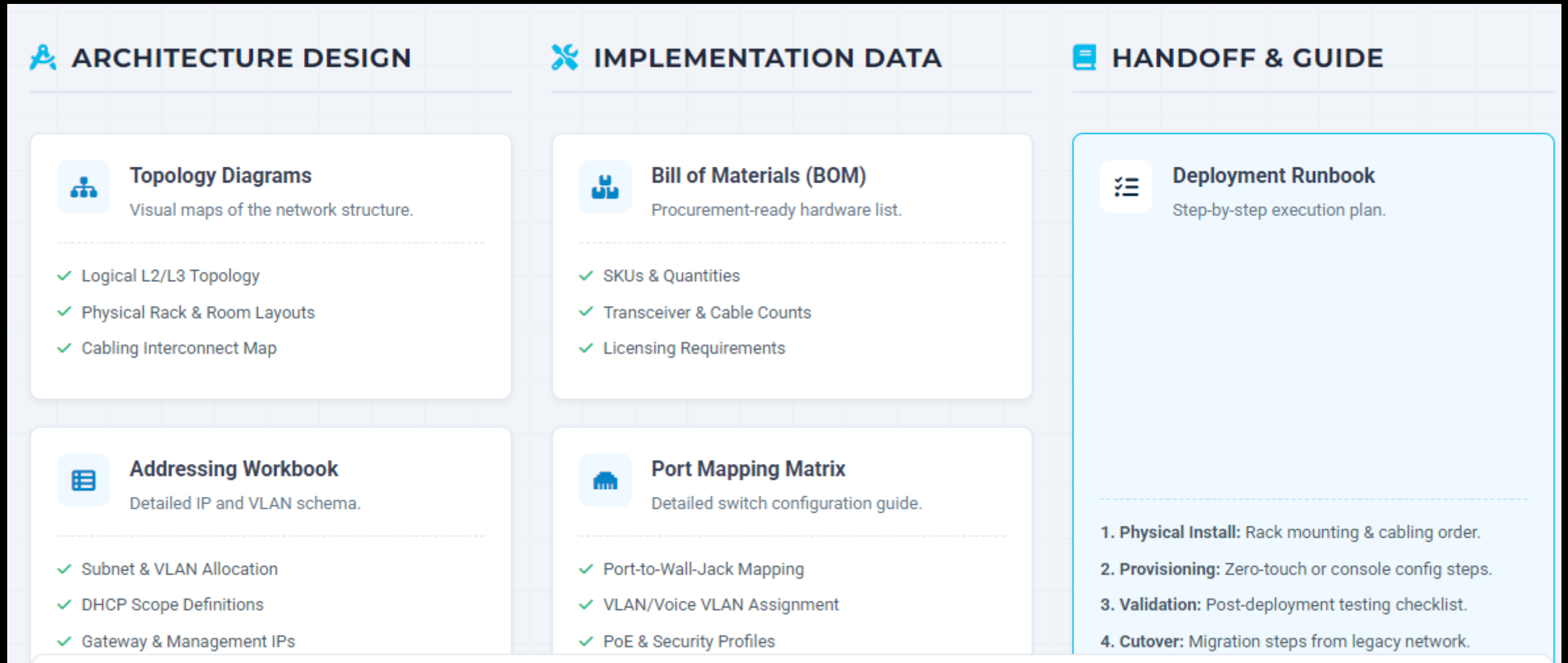
- Places firewalls at logical boundaries based on sensitivity zones.
- Generates control plane policing (CoPP) templates.
- Defines 802.1X authentication profiles for access ports.



Resilience Engineering

- Ensures dual-homing for critical paths and ECMP routing.
- Configures STP priorities or EVPN-VXLAN control planes.
- Maps failure domains to prevent single points of failure.

Final Output & Deployment Package



Technology

- AI & Intelligence Layer
 - ✓ Open-Source LLM / Local Deployment
 - ✓ Custom ML Models (TF-IDF + Logistic Regression, Entity Extraction Model)
 - ✓ RAG Architecture
 - ✓ Custom MCP Server (Model Orchestration Layer)
- Backend
 - ✓ Python
 - ✓ Flask (REST API Framework)
 - ✓ Node.js
- Frontend
 - ✓ React.js
 - ✓ Vite.js
 - ✓ Tailwind CSS
 - ✓ Axios
- Database
 - ✓ MongoDB
 - ✓ ChromaDB (In case of RAG) or Qdrant



References

- **HPE Aruba Networking Documentation (Primary References)**
- **Aruba Validated Solution Guides (VSG)** - Campus Network Design, Reference Architectures
Covers recommended topologies, segmentation, HA models, device roles, routing design, use cases.
- **AOS-CX Documentation Portal**
Includes switching concepts, VSX design principles, redundancy, VLAN/SVI design, routing behaviors.
- **AOS-10 Gateway / Mobility Documentation**
Guidance for tunneling/segmentation, secure zones, policy enforcement (only if needed for Exam/Security zones).
- **Aruba Central/CNX Workflows**
ZTP onboarding flow, Day-0/Day-1 config steps, device grouping & provisioning.
- **Product Datasheets & QuickSpecs**
Switches, gateways, and endpoint power requirements (PoE).
- **General Project References**
- Campus/Datacenter network hierarchy models
- Best-practice segmentation and addressing patterns
- Standardized cabling and port-mapping guidelines
- Architecture documentation patterns for enterprise networks

References

➤ Validated Solution Guide:

✓ <https://arubanetworking.hpe.com/techdocs/VSG/>

➤ AOS-CX Switches Models & Datasheet

✓ <https://www.hpe.com/us/en/aruba-cx-switches.html>

➤ Switches Buying Site:

✓ <https://buy.hpe.com/us/en/hpe-aruba-networking>

